

Smithsonian Fossil Palynomorph Detection and Classification

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1 Introduction

2 Background

3 Data

4 Proposed Models

We use the RF-DETR-based fossil palynomorph detection framework proposed by Abbas et al. [1] as the baseline for our study. Their existing whole-slide workflow detects fossil palynomorphs as a single object class using focus-stacked microscopy images, overlapping image tiles, RF-DETR inference, and duplicate-box merging.

Our objective is to extend this baseline from single-class localization to category-level identification. Specifically, we investigate whether fossil localization and category identification should be learned jointly within one object detection model or separated into detection and classification stages.

We therefore propose two modeling approaches. The first approach extends the baseline RF-DETR detector into a direct multiclass object detector that jointly predicts the location and category of each specimen. The second approach retains RF-DETR as a single-class detector and introduces a second-stage Swin-Tiny Transformer classifier to identify the category of each detected specimen.

These two approaches address complementary research questions. First, can localization and category identification be learned effectively within a single multiclass detector? Second, can a separate morphology-focused classifier improve category identification by operating directly on individual fossil crops?

To ensure a fair comparison, both approaches will use the same sponsor-confirmed target categories, source slides, slide-level training-validation-test split, focus-stacked inputs, tile generation procedure, and final held-out test set. Splitting will be performed at the source-slide level before tiling or augmentation in order to prevent visually related specimens from the same slide from appearing in multiple data partitions.

4.1 Model 1: Direct Multiclass RF-DETR

The first approach represents the most direct extension of the Abbas et al. baseline. Since RF-DETR already performs object localization and category prediction within a unified detection framework, extending its classification output from one palynomorph class to K target categories provides a straightforward end-to-end solution.

Our first proposed model therefore extends the existing single-class RF-DETR detector into a multiclass object detection model. Instead of predicting whether an object is simply a palynomorph, the model will jointly estimate the location of each fossil and its corresponding category.

Given an input image I , the detector produces a set of predictions

$$\hat{Y} = \left\{ (\hat{b}_i, \hat{c}_i, \hat{p}_i) \right\}_{i=1}^N,$$

where $\hat{b}_i$ represents the predicted bounding box of object i , $\hat{c}_i$ represents its predicted category, and $\hat{p}_i$ represents the associated confidence score.

The inference process for Model 1 is illustrated in Figure 1.

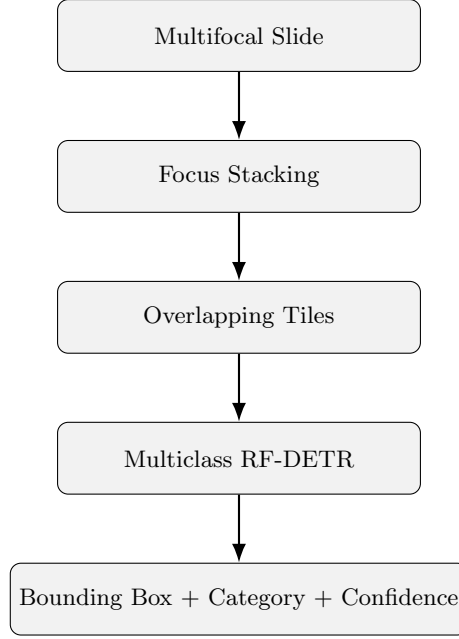


Figure 1: Inference pipeline for Model 1. Multifocal whole-slide images are converted into focus-stacked overlapping tiles, processed by a multiclass RF-DETR detector, and combined to produce specimen-level bounding boxes, predicted categories, and confidence scores.

Before training, each sponsor-approved fossil category will be mapped to a unique class identifier, and every annotated bounding box will be associated with both its category and source-slide identifier. The dataset will be partitioned into training, validation, and testing subsets at the slide level before any tiling or augmentation is performed. This prevents specimens from the same source slide from leaking across data partitions.

Focus-stacked tiles will then be generated using a fixed tile size and overlap strategy consistent with the baseline detection workflow. These preprocessing parameters will be kept identical across both proposed models so that differences in performance can be attributed primarily to the modeling strategy rather than to differences in input construction.

RF-DETR will be initialized from compatible pretrained weights, and its classification output will be modified to predict K target categories. The model will then be fine-tuned on the training set. Validation data will be used for checkpoint selection, confidence-threshold selection, and hyperparameter tuning. The final held-out test set will only be used after model selection is complete.

During inference, predictions from individual tiles will be mapped back to whole-slide coordinates. Duplicate detections created by overlapping tile boundaries will then be merged to produce the final set of bounding boxes, category labels, and confidence scores.

The primary advantage of this approach is its simplicity. A single model performs both localization and classification, allowing the detector to learn spatial and category-specific features jointly. However, rare or visually similar categories may be more difficult to learn within a joint detection objective. In addition, localization and classification errors are less easily separated during error analysis.

4.2 Model 2: RF-DETR with Swin-Tiny Classification

The second approach is designed to test whether separating localization from category identification provides an advantage over joint multiclass detection. This separation is motivated by the fact that fossil categories may differ through relatively subtle morphological characteristics that could be easier to learn from a centered specimen crop than from a larger detection tile.

Our second proposed model therefore separates fossil localization and category identification into two stages. In the first stage, RF-DETR detects all eligible fossil palynomorphs as a single object class. In the second stage, each detected object is cropped from the microscopy image and classified using a pretrained Swin-Tiny Transformer [2].

The two-stage inference process is illustrated in Figure 2.

4.2.1 Stage 1: Palynomorph Detection

For the detection stage, all eligible fossil annotations will first be collapsed into a single palynomorph class. RF-DETR will then be trained or fine-tuned using the same slide-level split, focus-stacked tiles, and duplicate-merging rules used for Model 1.

Because a fossil missed by the detector cannot be recovered by the classification stage, recall will be an important consideration when selecting the detector operating threshold. The goal of this stage is to generate reliable candidate bounding boxes that can subsequently be passed to the classifier.

4.2.2 Stage 2: Swin-Tiny Classification

The second stage focuses exclusively on classifying individual fossil specimens. Initial classifier training data will be generated using expert-annotated ground-

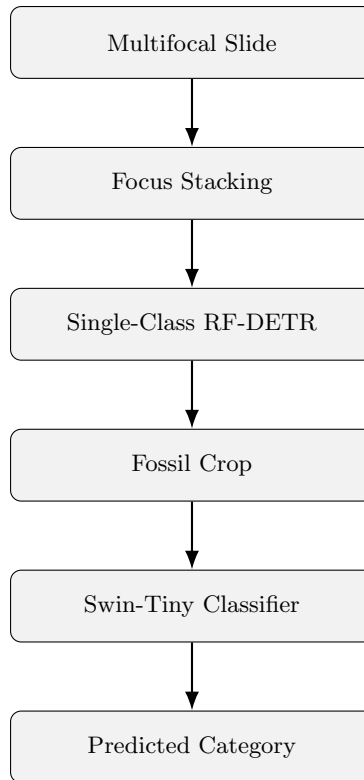


Figure 2: Inference pipeline for Model 2. RF-DETR first detects fossil paly-nomorphs as a single class. Each detected specimen is then cropped and passed to a Swin-Tiny classifier, which predicts its target category.

truth bounding boxes rather than RF-DETR predictions. This allows the classifier to be evaluated independently from detection errors.

Each fossil crop will include a fixed amount of surrounding context. The image will be padded to a square shape and resized to

$$224 \times 224$$

pixels without distorting the original aspect ratio of the specimen.

Swin-Tiny will be initialized from pretrained weights, and its final classification layer will be replaced with a new output layer corresponding to the K target fossil categories.

Swin Transformer uses a hierarchical architecture with shifted-window self-attention [2]. Rather than computing self-attention globally across the entire image at every layer, the model computes attention within local windows and shifts the window partition between successive Transformer blocks. This hierarchical structure enables the model to capture both local and progressively higher-level visual features while reducing the computational cost of global self-attention.

This architecture is relevant to the fossil classification task because differences between palynomorph categories may depend on fine-grained morphological characteristics such as aperture structure, wall structure, overall shape, and surface ornamentation. By operating on a centered fossil crop, Swin-Tiny can focus its representation on specimen morphology rather than simultaneously solving the localization problem.

During inference, RF-DETR first generates fossil detections. Each predicted bounding box is then converted into a crop using the same context, padding, and resizing procedure used during classifier training. Swin-Tiny predicts the category of each crop, and the classification result is attached to the original detection coordinates.

Detector confidence and classifier confidence will initially be retained separately. This allows detection uncertainty and classification uncertainty to be analyzed independently.

The major advantage of this two-stage approach is that the classifier can focus exclusively on the morphology of a centered fossil specimen. Detection and classification can also be evaluated independently, making error analysis more straightforward. However, errors from the detector propagate into the classification stage. Missed detections cannot be recovered, and inaccurate or incomplete bounding boxes may reduce classification performance.

4.3 Comparison of the Proposed Models

The two proposed models investigate different strategies for fossil palynomorph identification. Model 1 asks whether localization and category identification can be learned jointly by a single object detector. Model 2 asks whether separating localization from morphology-focused classification can improve category identification.

The main differences between the two approaches are summarized in Table 1.

Table 1: Comparison of the proposed model architectures.

Dimension	Model 1: Multiclass RF-DETR	Model 2: RF-DETR + Swin-Tiny
Strategy	Joint detection and classification	Separate detection and crop classification
Output	Bounding box, category, and confidence	Bounding box and detection confidence, followed by category and classifier confidence
Main Advantage	Simpler end-to-end inference	Clearer error analysis and specimen-focused classification
Main Dependency	Category label for every training bounding box	Reliable detector and category-labeled fossil crops
Main Failure Mode	Rare categories may be difficult to learn inside the joint objective	Missed or poor-quality detections reduce classifier performance
Research Question	Can localization and category identification be learned jointly?	Does a separate morphology-focused classifier improve identification?

4.4 Modeling Assumptions and Open Decisions

The proposed modeling framework currently relies on several assumptions that will be verified before final model training.

First, we assume that the sponsor-approved annotation categories can be mapped consistently to a fixed set of K target classes. The final treatment of uncertain, unknown, or excluded specimens will be determined in consultation with the project sponsors.

Second, we assume that the existing bounding-box annotations provide sufficiently accurate localization for both RF-DETR training and the construction of ground-truth crops for Swin-Tiny. Annotation quality will therefore be inspected before training, with particular attention to incorrect category labels, incomplete boxes, and inconsistent annotations.

Third, the proposed comparison assumes that both models receive equivalent training and evaluation data. The same source-slide split, preprocessing configuration, focus-stacking procedure, and evaluation set will therefore be used for both models.

Finally, for Model 2, we assume that a crop generated from the detected bounding box contains sufficient morphological information for category classi-

fication. The influence of crop quality will be tested by comparing Swin-Tiny performance on expert ground-truth crops and RF-DETR-predicted crops.

4.5 Model Evaluation

The two proposed models will be evaluated using the same held-out test set and compared against the original single-class RF-DETR baseline where applicable. The baseline comparison will first establish whether the palynomorph localization performance of the extended pipelines remains comparable to the existing detector. The two proposed models will then be compared on category-level and end-to-end performance.

Because the fossil category distribution may be imbalanced, overall accuracy alone will not be sufficient for evaluating model performance.

For the detection task, we will report Average Precision at an Intersection over Union threshold of 0.50,

$$AP_{50},$$

as well as mean Average Precision across IoU thresholds from 0.50 to 0.95,

$$AP_{50:95}.$$

Precision and recall will also be reported to measure whether fossil specimens are successfully detected and localized.

For category classification, we will evaluate Macro-F1, balanced accuracy, per-class precision, per-class recall, and the confusion matrix. These metrics provide a more informative evaluation when some fossil categories contain substantially fewer training examples than others.

End-to-end performance will additionally be evaluated using multiclass Average Precision and correctly classified specimen recall. These metrics measure whether a specimen is both successfully localized and assigned the correct category.

For Model 2, Swin-Tiny will first be evaluated on expert ground-truth crops. This experiment isolates classification performance from detection quality. The classifier will then be evaluated using RF-DETR-predicted crops to measure the performance of the deployable end-to-end system. A substantial performance difference between these two evaluations would indicate that detection or crop quality is a major bottleneck.

Finally, we will compare the computational cost of the two approaches using whole-slide inference time, peak memory usage, and the number of models required during inference.

To support reproducibility, all experiments will use documented dataset splits, preprocessing settings, model configurations, training hyperparameters, confidence thresholds, and random seeds. The same pipeline will also be designed to process previously unseen whole-slide images without requiring manual changes to the modeling procedure.

References

- [1] Abbas et al. *Scalable Detection of Fossil Palynomorphs in Multifocal Digital Microscopy Images*. arXiv preprint arXiv:2609.05323, 2026.
- [2] Z. Liu et al. *Swin Transformer: Hierarchical Vision Transformer Using Shifted Windows*. Proceedings of the IEEE/CVF International Conference on Computer Vision, pp. 10012–10022, 2021.